

Completing the small-prime congruences of OEIS A348410

A binary defect formula and a sharp
denominator-12 theorem

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Abstract

Let $a(n) = [x^n]((1-x)(1-x^2))^{-n}$. An OEIS comment of Peter Bala conjectures the cubic congruence $a(mp^r) \equiv a(mp^{r-1}) \pmod{p^{3r}}$ for primes $p \geq 5$. A source audit located a prior elementary proof candidate for that assertion and a two-parameter extension; we therefore do not present the odd-prime result as a new resolution. Instead, we investigate the binary case explicitly left untreated in that source.

For $A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}$, with arbitrary integer parameters, we prove an explicit first-defect congruence modulo 2^{3r-2} . It gives a sharp universal modulus 2^{3r-3} , improved to 2^{3r-2} when $\alpha\beta$ is even. Combining this with a self-contained odd-prime proof gives $12n^{-3} \sum_{d|n} \mu(d)a(n/d) \in \mathbb{Z}$ for every $n \geq 1$. The constant 12 is the smallest possible common denominator. We also obtain a primitive-necklace divisibility theorem, explain the failure of an overbroad periodic-coefficient argument, derive algebraic generating functions and two correction terms in the asymptotic expansion, and provide exact reproducible checks. Mathematical completeness of the written proofs and global priority are separate matters: no claim of exhaustive novelty verification or external peer review is made.

Archive contents. This article in L^AT_EX and PDF; three verification programs; exact CSV tables; test reports; build instructions; a source and priority audit; and a draft OEIS comment. No computational conjecture is used as a premise of a theorem.

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1 The problem, the source boundary, and the results

1.1 Why the target changed during the investigation

The sequence A348410 counts nonnegative integer solutions of

$$n = \sum_{i=1}^n (u_i + 2v_i). \quad (1.1)$$

Equivalently, n indistinguishable objects are distributed among n labelled white baskets and n labelled black baskets, and the black occupancies are even. We set $a(0) = 1$. The OEIS entry records the coefficient formula

$$a(n) = [x^n] \frac{1}{(1-x)^n(1-x^2)^n} \quad (1.2)$$

and, in a comment dated 21 February 2022, Bala's conjecture

$$a(mp^r) - a(mp^{r-1}) \in p^{3r}\mathbb{Z} \quad (p \geq 5 \text{ prime, } m, r \geq 1). \quad (1.3)$$

The live entry still labels this a conjecture in the version inspected for this article [1].

That label is not a reliable certificate of present-day openness. In particular, the 2026 articles of Niu and Prodinger already discuss the algebraic generating function [5, 6]. More importantly for this investigation, a public repository note, *A cubic tower for a two-parameter coefficient family*, contains an elementary proof candidate for (1.3), for its full integer-parameter analogue, and for a 3^{3r-1} ternary bound [3]. Its parameter convention is $[x^n]((1+x)^u(1-x)^v)^n$; ours corresponds to $(u, v) = (-\beta, -\alpha)$. The full odd-prime parameter conjecture is also stated on A352373 [2]. The repository source explicitly leaves the binary case without an assertion.

The research question used here is therefore the following more precise completion problem, not a claim that the odd-prime conjecture had no prior proof.

Problem 1.1 (Binary completion). For the integer-parameter family

$$A_{\alpha,\beta}(n) = [x^n](1-x)^{-\alpha n}(1+x)^{-\beta n}, \quad \alpha, \beta \in \mathbb{Z}, \quad (1.4)$$

does cubic growth of the divisibility exponent persist at $p = 2$, with a fixed loss independent of the level? Can one determine the first possible binary defect and obtain an optimal common denominator for the cubic Möbius transform?

The sources checked did not supply the binary formula or the denominator conclusions below. This is a statement about the checked source boundary, not an exhaustive claim about all mathematical literature. The odd-prime result is credited to the prior proof candidate and reproved here to make the new completion independent of an external unreviewed argument.

1.2 Main theorems

All binomial coefficients with an integer upper argument use the generalized convention

$$\binom{z}{k} = \frac{z(z-1)\cdots(z-k+1)}{k!}, \quad k \geq 0. \quad (1.5)$$

In particular, this is an integer when z is a negative integer.

Theorem 1.2 (First binary defect). *Let $\alpha, \beta \in \mathbb{Z}$, let $m \geq 1$ be odd, and let $r \geq 2$. Then*

$$\begin{aligned} & A_{\alpha,\beta}(m2^r) - A_{\alpha,\beta}(m2^{r-1}) \\ & \equiv 2^{3r-3} \alpha \beta \binom{(\alpha + \beta + 1)m - 1}{m - 1} \pmod{2^{3r-2}}. \end{aligned} \quad (1.6)$$

Consequently the difference is always divisible by 2^{3r-3} , and is divisible by 2^{3r-2} whenever $\alpha\beta$ is even. If both parameters are odd and $m = 1$, then its valuation is exactly $3r - 3$, at every $r \geq 2$.

The first lift $r = 1$ is deliberately not included in the displayed defect formula. It always satisfies the elementary congruence modulo 2 proved below. The distinction prevents a common error: extrapolating a later-level binary estimate to the first lift.

Theorem 1.3 (All primes for the basket sequence). *For every prime p and all $m, r \geq 1$,*

$$v_p(a(mp^r) - a(mp^{r-1})) \geq 3r - v_p(12). \quad (1.7)$$

Thus the moduli are p^{3r} for $p \geq 5$, 3^{3r-1} for $p = 3$, and 2^{3r-2} for $p = 2$. Each of these uniform losses is necessary.

For a zero difference, its valuation is interpreted as $+\infty$.

Theorem 1.4 (Sharp common denominators). *Define*

$$B_{\alpha,\beta}(n) = \frac{1}{n^3} \sum_{d|n} \mu(d) A_{\alpha,\beta}(n/d), \quad B(n) = B_{2,1}(n). \quad (1.8)$$

Then $24B_{\alpha,\beta}(n) \in \mathbb{Z}$ for all integer parameters and all $n \geq 1$. If $\alpha\beta$ is even, $12B_{\alpha,\beta}(n) \in \mathbb{Z}$. In particular,

$$\boxed{\frac{12}{n^3} \sum_{d|n} \mu(d) a(n/d) \in \mathbb{Z}_{\geq 0} \quad (n \geq 1).} \quad (1.9)$$

The constant 12 is minimal for the single sequence $a(n)$. The constant 24 is minimal for the unrestricted integer-parameter family.

For the combinatorial interpretation, let P_n be the number of primitive cyclic words of length n in the weighted alphabet of pairs (u, v) of nonnegative integers, where (u, v) has weight $u + 2v$, and the total word weight is n . “Primitive” means that no shorter word is repeated to form it. Cyclic rotations are identified.

Corollary 1.5 (Primitive-orbit divisibility). *For every $n \geq 1$,*

$$P_n = \frac{1}{n} \sum_{d|n} \mu(d) a(n/d), \quad \frac{n^2}{\gcd(n^2, 12)} \mid P_n. \quad (1.10)$$

1.3 What is proved, and what is not claimed

The binary theorem and denominator theorem have complete elementary proofs in Sections 4 and 5. The odd-prime proof in Section 3 is included as a self-contained alternative to the prior repository argument. The algebraic generating function and leading asymptotic are not claimed as new: both are already documented in the OEIS and related literature. The correction coefficients and the general Gaussian-moment expression are derived explicitly here,

without a priority claim. A stronger exact valuation pattern at powers of two is labelled as conjectural in Section 10.

The proofs have been checked algebraically and by independent exact computations, but have not been externally refereed or formalized in Lean. The code is a reproducibility aid, not a substitute for the arguments.

2 Counting formulas and exact coefficient computation

2.1 A positive binomial sum

A single white basket contributes $(1 - x)^{-1}$ and a black basket contributes $(1 - x^2)^{-1}$. Multiplying over the labelled baskets proves (1.2). Expanding the two factors separately gives, for $n \geq 1$,

$$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{2n - 2k - 1}{n - 2k} \binom{n + k - 1}{k}. \quad (2.1)$$

The first terms are

$$1, 1, 5, 19, 85, 376, 1715, 7890, 36693, 171820, 809380.$$

This positive sum gives an independent check on the faster coefficient recurrence below. It also makes integrality and nonnegativity immediate for the target sequence.

For arbitrary integer parameters, generalized binomial expansion shows that

$$A_{\alpha,\beta}(n) = \sum_{k=0}^n (-1)^k \binom{\beta n + k - 1}{k} \binom{\alpha n + n - k - 1}{n - k} \quad (n \geq 1). \quad (2.2)$$

This formula includes zero and negative parameters without any change of convention. We put $A_{\alpha,\beta}(0) = 1$.

2.2 A two-step recurrence in the coefficient index

Fix n and write

$$F(x) = (1 - x)^{-\alpha n} (1 + x)^{-\beta n} = \sum_{k \geq 0} f_k x^k.$$

Logarithmic differentiation gives

$$(1 - x^2)F'(x) = n((\alpha - \beta) + (\alpha + \beta)x)F(x).$$

Consequently, with $f_{-1} = 0$ and $f_0 = 1$,

$$k f_k = n(\alpha - \beta) f_{k-1} + (n(\alpha + \beta) + k - 2) f_{k-2}, \quad k \geq 1. \quad (2.3)$$

The desired output is f_n . This is a recurrence in k with n fixed, not a recurrence relating successive diagonal values $A_{\alpha,\beta}(n)$.

The algorithm uses $O(n)$ exact arithmetic operations and two live integer coefficients. The integers grow with n , so this is not a claim of linear bit complexity. Every division by k is checked for exactness in the supplied implementation. Large datasets are computed without floating-point arithmetic.

2.3 Small-prime boundaries visible in the first terms

For the target sequence,

$$\begin{aligned} a(2) - a(1) &= 4, \\ a(3) - a(1) &= 18 = 2 \cdot 3^2, \\ a(4) - a(2) &= 80 = 5 \cdot 2^4, \\ a(5) - a(1) &= 375 = 3 \cdot 5^3, \\ a(7) - a(1) &= 7889 = 23 \cdot 7^3. \end{aligned}$$

These examples already show that $p = 2$ and $p = 3$ cannot simply be inserted into Bala's stated modulus. They also prove the necessity of the fixed losses in Theorem 1.3: at $p = 2$ use level $r = 2$, and at $p = 3$ use level $r = 1$. At $p = 5$ there is no extra uniform power available.

3 Local arithmetic and the odd-prime theorem

This section proves the odd-prime result independently. It overlaps in scope with the prior coefficient-framing theorem [3]; the direct coefficient-valuation argument below avoids dependence on the general rational-framing statement discussed in Section 7.

3.1 The local ring and a coefficient estimate

For a prime p , let

$$R_p = \mathbb{Z}_{(p)} = \{u/v \in \mathbb{Q} : p \nmid v\}.$$

Congruence modulo p^h in this ring means that the difference has valuation at least h . A formal power-series congruence is always coefficientwise. All exponentials and logarithms are formal over $\mathbb{Q}$; at a fixed degree only finitely many terms contribute.

Set

$$\Phi(x) = (1 - x)^{-\alpha}(1 + x)^{-\beta}, \quad L(x) = \log \Phi(x) = \sum_{j \geq 1} \frac{c_j}{j} x^j, \quad c_j = \alpha + \beta(-1)^j. \quad (3.1)$$

Lemma 3.1 (Valuation of a coefficient). *Let $M \geq 1$ and write $g_j = [x^j]\Phi(x)^M$. Then $g_j \in \mathbb{Z}$, $g_0 = 1$, and for $j \geq 1$,*

$$v_p(g_j) \geq \max\{0, v_p(M) - v_p(j)\}. \quad (3.2)$$

Proof. Integrality follows from generalized binomial expansion. Differentiating $\Phi(x)^M$ and multiplying by x yields

$$jg_j = M \sum_{i=1}^j c_i g_{j-i}.$$

The sum on the right is an integer. Thus $v_p(g_j) \geq v_p(M) - v_p(j)$, and integrality gives the additional lower bound zero. $\square$

3.2 A reciprocal-square cancellation

For odd p , put

$$\varepsilon_p = \begin{cases} 1, & p = 3, \\ 0, & p \geq 5. \end{cases}$$

Lemma 3.2 (Reduced reciprocal squares). *If $s > 0$, $p \mid s$, and $t = v_p(s)$, then*

$$H_p(s) = \sum_{\substack{1 \leq j < s \\ p \nmid j}} \frac{1}{j^2} \in p^{t-\varepsilon_p} R_p. \quad (3.3)$$

Proof. Reduce denominators modulo p^t and partition the summation interval into complete blocks of length p^t . It is enough to consider $U_t = \sum_{u \in (\mathbb{Z}/p^t\mathbb{Z})^\times} u^{-2}$.

For $p \geq 5$, multiplication by 2 permutes the units and gives $U_t \equiv 2^{-2}U_t \pmod{p^t}$. Hence $3U_t \equiv 0 \pmod{p^t}$; since 3 is invertible, $U_t \equiv 0$.

For $p = 3$, induction proves $U_t \equiv 0 \pmod{3^{t-1}}$. The case $t = 1$ has modulus 1. For $t \geq 2$, each unit modulo 3^{t-1} has three lifts, so $U_t \equiv 3U_{t-1} \pmod{3^{t-1}}$. The induction hypothesis finishes the proof. The same bounds hold for the full block sum. $\square$

For odd p , parity is preserved by multiplication by p , so $c_{pj} = c_j$. The reduced logarithm

$$D_p(x) = L(x) - \frac{1}{p}L(x^p) = \sum_{\substack{j \geq 1 \\ p \nmid j}} \frac{c_j}{j} x^j \quad (3.4)$$

belongs to $xR_p[[x]]$ and has no monomial whose exponent is divisible by p .

Lemma 3.3 (Quadratic-tail estimate). *If $s > 0$ is divisible by the odd prime p , then*

$$T_p(s) := [x^s]D_p(x)^2 \in p^{v_p(s)-\varepsilon_p} R_p. \quad (3.5)$$

Proof. Write $t = v_p(s)$. Since $p \mid s$, both j and $s - j$ are units whenever $p \nmid j$. Thus

$$T_p(s) = \sum_{\substack{1 \leq j < s \\ p \nmid j}} \frac{c_j c_{s-j}}{j(s-j)} \equiv - \sum_{\substack{1 \leq j < s \\ p \nmid j}} \frac{c_j c_{s-j}}{j^2} \pmod{p^t}.$$

If s is odd, the weight $c_j c_{s-j} = \alpha^2 - \beta^2$ is constant, so Lemma 3.2 applies directly.

If s is even, the weights are $(\alpha + \beta)^2$ for even j and $(\alpha - \beta)^2$ for odd j . The weighted reciprocal-square sum is exactly

$$(\alpha - \beta)^2 H_p(s) + \alpha\beta H_p(s/2).$$

Indeed, the excess even weight is $4\alpha\beta$, while replacing j by $2k$ divides the reciprocal square by 4. Since p is odd, $v_p(s/2) = t$. Applying Lemma 3.2 to both terms proves the claim. $\square$

3.3 Proof of the full odd-prime tower

Theorem 3.4 (Odd-prime coefficient tower). *For every $\alpha, \beta \in \mathbb{Z}$, odd prime p , and $N > 0$ divisible by p , write $r = v_p(N)$. Then*

$$A_{\alpha,\beta}(N) - A_{\alpha,\beta}(N/p) \in p^{3r-\varepsilon_p} \mathbb{Z}. \quad (3.6)$$

Proof. Put $N = pM$. From (3.4) one obtains the exact identity

$$\Phi(x)^N = \Phi(x^p)^M \exp(ND_p(x)). \quad (3.7)$$

The factorial valuation bounds imply

$$\exp(ND_p) \equiv 1 + ND_p + \frac{N^2}{2}D_p^2 \pmod{p^{3r-\varepsilon_p}R_p[[x]]}. \quad (3.8)$$

Here are explicit bounds, including the small exponential degrees. For $p \geq 5$ and $k = 3$, $v_p(k!) = 0$. For $k \geq 4$, $v_p(k!) \leq (k-1)/4 \leq k-3$, so $kr - v_p(k!) \geq 3r$. For $p = 3$ and $k \geq 3$, $v_3(k!) \leq (k-1)/2 \leq k-2$, and $kr - v_3(k!) \geq 3r-1$. These estimates justify every discarded term of the formal exponential.

Extract degree N . The constant term of the exponential gives $A_{\alpha,\beta}(M)$. The linear contribution is zero: $\Phi(x^p)^M$ is supported on exponents divisible by p , whereas D_p is supported away from them. Therefore

$$A_{\alpha,\beta}(N) - A_{\alpha,\beta}(M) \equiv \frac{N^2}{2}[x^N]\Phi(x^p)^M D_p(x)^2 \pmod{p^{3r-\varepsilon_p}}. \quad (3.9)$$

Writing $g_j = [x^j]\Phi(x)^M$, the remaining coefficient is

$$\sum_{j=0}^{M-1} g_j T_p(p(M-j)). \quad (3.10)$$

We claim that each summand belongs to $p^{r-\varepsilon_p}R_p$. For $j = 0$, this follows directly from Lemma 3.3 at $s = N$. For $j > 0$ with $v_p(j) < r-1$, the equality $v_p(M) = r-1$ gives $v_p(M-j) = v_p(j)$. Lemmas 3.1 and 3.3 then supply, respectively,

$$r-1-v_p(j) \quad \text{and} \quad 1+v_p(j)-\varepsilon_p,$$

whose sum is $r-\varepsilon_p$. For $v_p(j) \geq r-1$, we have $v_p(M-j) \geq r-1$, and the quadratic-tail bound alone supplies $r-\varepsilon_p$.

Finally, $v_p(N^2/2) = 2r$ because p is odd. Inserting the valuation of (3.10) into (3.9) proves (3.6). The difference is an ordinary integer, so its local divisibility is ordinary integer divisibility. $\square$

For $N = mp^r$ with $p \mid m$, the exact valuation $v_p(N)$ is larger than the displayed level. The theorem therefore implies the usual formulation without any coprimality restriction on m .

4 The binary defect: proof and sharpness

At $p = 2$, the equality $c_{pj} = c_j$ used in (3.4) fails. The correct replacement is a reflection argument. It separates the even and odd parts before any exponential is truncated.

4.1 An exact odd-harmonic normalization

Work in $R_2 = \mathbb{Z}_{(2)}$ and define

$$U(x) = \sum_{\substack{j \geq 1 \\ j \text{ odd}}} \frac{x^j}{j} = \frac{1}{2} \log \frac{1+x}{1-x}. \quad (4.1)$$

The factor $1/2$ in the logarithmic expression is harmless: the displayed odd-power series has coefficients in R_2 .

Lemma 4.1 (Odd harmonic sums are exactly normalized). *For every positive even T , let*

$$h(T) = \sum_{\substack{1 \leq j < T \\ j \text{ odd}}} \frac{1}{j}.$$

Then

$$v_2(h(T)) = 2v_2(T) - 2. \quad (4.2)$$

In particular, $h(2j)/j^2$ is an odd element of R_2 for every $j \geq 1$.

Proof. Write $T = 2^e q$ with q odd. First,

$$\sum_{\substack{1 \leq u < 2^e \\ u \text{ odd}}} u^{-2} \equiv 2^{e-1} \pmod{2^e}. \quad (4.3)$$

For $e = 1$ this is immediate. For $e \geq 2$, the two lifts u and $u + 2^{e-1}$ of an odd residue modulo 2^{e-1} have equal inverse squares modulo 2^e . Indeed their squares differ by a multiple of 2^e . Thus the sum at level e is twice the sum at level $e - 1$ modulo 2^e , which proves (4.3) by induction.

Partitioning the interval $1 \leq j < T$ into q blocks now shows that $\sum_{j \text{ odd}} j^{-2} \equiv 2^{e-1} \pmod{2^e}$. Pair j with $T - j$ to obtain the exact identity

$$h(T) = \frac{T}{2} \sum_{\substack{1 \leq j < T \\ j \text{ odd}}} \frac{1}{j(T-j)}.$$

Modulo 2^e , the sum on the right is the negative of the reciprocal-square sum. It consequently has valuation exactly $e - 1$. The prefactor $T/2$ has valuation $e - 1$ as well, proving (4.2). $\square$

The even power series $U(x)^2$ satisfies

$$[x^{2j}]U(x)^2 = \sum_{\substack{1 \leq k < 2j \\ k \text{ odd}}} \frac{1}{k(2j-k)} = \frac{h(2j)}{j}.$$

Define

$$K(t) = \sum_{j \geq 1} \frac{h(2j)}{j^2} t^j \in tR_2[[t]], \quad V(t) = U(\sqrt{t})^2. \quad (4.4)$$

Here $U(\sqrt{t})^2$ denotes the integer-exponent series obtained from the even series $U(x)^2$ by replacing x^2 with t ; no choice of an analytic square root is involved. The identities we need are

$$V(t) = tK'(t), \quad U(t)^2 = \frac{t}{2} \frac{d}{dt} K(t^2), \quad K(t) \equiv \frac{t}{1-t} \pmod{2}. \quad (4.5)$$

The first two follow by coefficient comparison. The last follows from the oddness assertion of Lemma 4.1. This final residue, not just a divisibility bound, will determine the leading binary digit.

4.2 Separating the two coefficient problems

Set

$$S = \alpha + \beta, \quad \delta = \alpha - \beta.$$

Let M be even, put $s = v_2(M) \geq 1$, and define

$$g(t) = (1 - t)^{-SM}.$$

The logarithmic identities

$$\begin{aligned} \Phi(x)^{2M} &= (1 - x^2)^{-SM} \exp(2\delta MU(x)), \\ \Phi(t)^M &= g(t) \exp(-2\beta MU(t)) \end{aligned}$$

are exact. Only the even part of the first exponential contributes to degree $2M$. Consequently

$$\begin{aligned} A_{\alpha,\beta}(2M) &= [t^M]g(t) \cosh(2\delta MU(\sqrt{t})), \\ A_{\alpha,\beta}(M) &= [t^M]g(t) \exp(-2\beta MU(t)). \end{aligned} \quad (4.6)$$

The hyperbolic cosine is interpreted as its even formal Taylor series, so every exponent of t in (4.6) is an integer.

Since $v_2(k!) \leq k - 1$, for $k \geq 3$ we have

$$v_2\left(\frac{(2M)^k}{k!}\right) \geq ks + 1 \geq 3s + 1.$$

The same estimate applies to all even degrees $k \geq 4$ in the cosine. Thus modulo 2^{3s+1} we need only three coefficient expressions:

$$I_1 = [t^M]gU, \quad I_2 = [t^M]gU^2, \quad J = [t^M]gV. \quad (4.7)$$

Writing $\Delta = A_{\alpha,\beta}(2M) - A_{\alpha,\beta}(M)$, expansion of (4.6) gives

$$\Delta \equiv 2\beta MI_1 + 2\delta^2 M^2 J - 2\beta^2 M^2 I_2 \pmod{2^{3s+1}}. \quad (4.8)$$

4.3 Reflection converts the linear term into a quadratic one

The apparent obstacle in (4.8) is the linear term I_1 . Because M is even and U is odd,

$$[t^M]g(-t)U(t) = -I_1.$$

Also

$$\frac{g(-t)}{g(t)} = \left(\frac{1-t}{1+t}\right)^{SM} = \exp(-2SMU(t)).$$

Combining these identities yields

$$I_1 = \frac{1}{2}[t^M]g(t)U(t) (1 - \exp(-2SMU(t))). \quad (4.9)$$

The first exponential term in (4.9) is SMI_2 . Every remaining term has a scalar coefficient with valuation at least $2s$: for $k \geq 2$,

$$v_2\left(\frac{(2SM)^k}{2k!}\right) \geq k(s+1) - 1 - v_2(k!) \geq ks \geq 2s.$$

Hence

$$I_1 \equiv SMI_2 \pmod{2^{2s}R_2}. \quad (4.10)$$

Multiplication by $2\beta M$ makes the error divisible by 2^{3s+1} . Substituting in (4.8) and using $\beta S - \beta^2 = \alpha\beta$ therefore gives

$$\Delta \equiv 2M^2(\alpha\beta I_2 + \delta^2 J) \pmod{2^{3s+1}}. \quad (4.11)$$

This reflection is the step that replaces the odd-prime support cancellation. It is valid because M is even; this is exactly why the first binary lift was separated from the theorem.

4.4 Formal integration by parts supplies the third power

For any $K_0(t) \in R_2[[t]]$, coefficient differentiation gives

$$[t^M]g(t)tK_0'(t) = M[t^M]g(t)K_0(t) \left(1 - \frac{St}{1-t}\right). \quad (4.12)$$

Indeed, $[t^M]t(gK_0)' = M[t^M]gK_0$ and $tg'/g = SMt/(1-t)$.

Using (4.5) in (4.12) shows

$$J \in MR_2, \quad I_2 = \frac{M}{2}C, \quad (4.13)$$

where

$$C = [t^M](1-t)^{-SM}K(t^2) \left(1 - \frac{St}{1-t}\right) \in R_2. \quad (4.14)$$

Thus the J term in (4.11) is already zero modulo 2^{3s+1} , and we obtain the particularly useful congruence

$$\Delta \equiv \alpha\beta M^3 C \pmod{2^{3s+1}}. \quad (4.15)$$

It immediately proves the universal valuation bound $3s$ and its one-bit improvement when $\alpha\beta$ is even.

It remains to compute C modulo 2 when both parameters are odd. In this case S is even. Write $M = 2^s m$ with m odd. Over $\mathbb{F}_2[[t]]$, generalized integer powers and the Frobenius identity give

$$(1-t)^{-SM} \equiv (1-t^{2^s})^{-Sm}, \quad 1 - \frac{St}{1-t} \equiv 1,$$

while $K(t^2) \equiv t^2/(1-t^2)$. Since $2^s \geq 2$, extraction of t^M then gives

$$\begin{aligned} C &\equiv \sum_{j=0}^{m-1} \binom{Sm+j-1}{j} \\ &= \binom{(S+1)m-1}{m-1} \pmod{2}. \end{aligned} \quad (4.16)$$

The last identity is the generalized hockey-stick identity, obtained by multiplying $(1-z)^{-Sm}$ by $(1-z)^{-1}$ and extracting degree $m-1$. It holds for every integer S , including negative values.

If $\alpha\beta$ is even, both sides of the desired leading-bit formula vanish regardless of C . If $\alpha\beta$ is odd, insert (4.16) into (4.15). Finally, $M^3 = 2^{3s}m^3 \equiv 2^{3s} \pmod{2^{3s+1}}$. Taking $s = r-1$ proves Theorem 1.2 in full. For $m = 1$ the binomial coefficient in (1.6) equals 1, so odd parameters give an odd leading quotient and exact valuation $3r-3$.

4.5 First lift, optimal losses, and the all-prime statement

For any $\Phi \in 1 + x\mathbb{Z}[[x]]$,

$$\Phi(x)^2 \equiv \Phi(x^2) \pmod{2}.$$

Raising this to m and extracting degree $2m$ proves

$$A_{\alpha,\beta}(2m) \equiv A_{\alpha,\beta}(m) \pmod{2}. \quad (4.17)$$

For $(\alpha, \beta) = (2, 1)$, the improved binary bound of Theorem 1.2 is $3r - 2$ for $r \geq 2$, and (4.17) supplies exactly the claimed lower bound 1 at $r = 1$. If m is even, apply the theorem at the larger exact binary level. Together with Theorem 3.4, this proves Theorem 1.3. The examples in Section 2 prove its uniform sharpness.

There is a distinction between uniform sharpness and eventual sharpness. For the target sequence, equality occurs at $r = 2$, but later powers of two show more divisibility. The theorem does not assert equality at every level. For the unrestricted family, in contrast, odd parameters and $m = 1$ attain the general bound $3r - 3$ at every level $r \geq 2$.

5 Möbius transforms, common denominators, and cyclic words

5.1 From a prime-power tower to a denominator theorem

For any sequence $A(n)$, set

$$T_A(n) = \sum_{d|n} \mu(d) A(n/d).$$

If $n = p^r n_0$ with $p \nmid n_0$, separating divisors according as they contain p gives the exact identity

$$T_A(n) = \sum_{d|n_0} \mu(d) \left(A(p^r n_0/d) - A(p^{r-1} n_0/d) \right). \quad (5.1)$$

Only squarefree divisors matter, so there are no terms containing a higher power of p in the divisor itself.

Apply (5.1) to the basket sequence. Theorem 1.3 implies, for every prime $p \mid n$,

$$v_p(T_a(n)) \geq 3v_p(n) - v_p(12).$$

Therefore $n^3 \mid 12T_a(n)$, proving the integrality part of (1.9).

For the general family, the odd-prime bounds are unchanged and the binary bound loses at most three powers. At level one the needed general estimate is merely valuation at least zero. Consequently

$$n^3 \mid 24T_{A_{\alpha,\beta}}(n).$$

If $\alpha\beta$ is even, the improved binary estimate and (4.17) reduce 24 to 12. This proves all the integrality assertions of Theorem 1.4.

The constants cannot be reduced. For the target sequence,

$$B(3) = \frac{19-1}{27} = \frac{2}{3}, \quad B(4) = \frac{85-5}{64} = \frac{5}{4}. \quad (5.2)$$

Every integer clearing all denominators must thus be divisible by both 3 and 4. Their least common multiple is 12. For the unrestricted family, $(\alpha, \beta) = (1, 1)$ gives

$$B_{1,1}(4) = \frac{10-2}{64} = \frac{1}{8},$$

and the target example at $n = 3$ forces a factor 3, so a universal clearing integer must be divisible by 24. Even the single parameter pair $(1, 3)$ requires 24: its values at $n = 3$ and $n = 4$ are $-8/3$ and $59/8$.

5.2 Primitive cyclic words and positivity

Let $\mathcal{A} = \{(u, v) : u, v \in \mathbb{Z}_{\geq 0}\}$, with weight $u + 2v$. A word of length n and total weight n corresponds exactly to a solution of (1.1). Although the alphabet is infinite, the weight constraint permits only finitely many letters at each fixed n .

A word of length n whose least period is $d \mid n$ is the repetition of a primitive block of length d . Its total weight is n precisely when the block weight is d . A primitive cyclic orbit of such blocks has exactly d distinct linear representatives. Therefore

$$a(n) = \sum_{d \mid n} dP_d. \quad (5.3)$$

Möbius inversion proves the first formula of (1.10) and also proves $P_n \geq 0$. Since $B(n) = P_n/n^2$, nonnegativity in Theorem 1.4 follows. Integrality of $12P_n/n^2$ is equivalent to the second assertion of (1.10).

The first values illustrate that this is stronger than the ordinary integrality of a necklace count:

n	P_n	$B(n) = P_n/n^2$	$12B(n)$
1	1	1	12
2	2	1/2	6
3	6	2/3	8
4	20	5/4	15
5	75	3	36
6	282	47/6	94
7	1127	23	276
8	4576	143/2	858

For example, $P_7 = 49 \cdot 23$, and P_6 is divisible by $36/\gcd(36, 12) = 3$. The individual counts need not be multiples of n^2 ; small-prime denominators are genuine.

5.3 Lambert series and an Euler product

Let $H(z) = \sum_{n \geq 0} a(n)z^n$. Equation (5.3) gives the formal Lambert expansion

$$H(z) - 1 = \sum_{n \geq 1} nP_n \frac{z^n}{1 - z^n} = \sum_{n \geq 1} n^3 B(n) \frac{z^n}{1 - z^n}. \quad (5.4)$$

All operations are valid formally because only finitely many divisors contribute to a fixed coefficient. The denominator theorem can be stated as follows: the cubic Lambert coefficients of $12(H - 1)$ are nonnegative integers, and 12 is the least positive integer with this property.

A related Euler product will follow from the Lagrange series in the next section:

$$\frac{y(z)}{z} = \prod_{n \geq 1} (1 - z^n)^{-P_n}. \quad (5.5)$$

To verify it directly, take formal logarithms. The coefficient of z^k on the right is $k^{-1} \sum_{n \mid k} nP_n = a(k)/k$, exactly the logarithm on the left derived below.

6 Algebraic generating functions

The main generating-function identities are established facts for A348410 [1, 5, 6]. We include a short derivation because it connects the coefficient-power definition, the Euler product, and the asymptotic analysis.

6.1 Lagrange reversion for the full parameter family

Let $y(z)$ be the unique series in $z\mathbb{Z}[[z]]$ satisfying

$$y = z\Phi(y). \quad (6.1)$$

Existence and uniqueness follow recursively from $\Phi(0) = 1$. Write $H_{\alpha,\beta}(z) = \sum_{n \geq 0} A_{\alpha,\beta}(n)z^n$. Lagrange inversion, applied to $\log \Phi(y)$, gives

$$\begin{aligned} [z^n] \log \Phi(y(z)) &= \frac{1}{n} [x^{n-1}] L'(x) \Phi(x)^n \\ &= \frac{1}{n^2} [x^{n-1}] \frac{d}{dx} \Phi(x)^n = \frac{A_{\alpha,\beta}(n)}{n}. \end{aligned}$$

Since $y/z = \Phi(y)$, it follows that

$$\log \frac{y(z)}{z} = \sum_{n \geq 1} \frac{A_{\alpha,\beta}(n)}{n} z^n. \quad (6.2)$$

Differentiating and using (6.1) yields

$$H_{\alpha,\beta}(z) = \frac{zy'(z)}{y(z)} = \frac{1 - y^2}{1 - (\alpha - \beta)y - (\alpha + \beta + 1)y^2}. \quad (6.3)$$

For integer parameters, Φ is rational. Thus (6.1) becomes a polynomial equation after clearing denominators, and $H_{\alpha,\beta}$ is algebraic.

6.2 The target quartic

For $(\alpha, \beta) = (2, 1)$,

$$z = y(1 - y)^2(1 + y), \quad H(z) = \frac{1 - y^2}{1 - y - 4y^2}. \quad (6.4)$$

Eliminating y gives

$$\begin{aligned} z^2(4H - 1)^4 + zH(107H^3 - 107H^2 + 36H - 4) \\ + 32H^3(1 - H) = 0. \end{aligned} \quad (6.5)$$

Substitution of (6.4) verifies the identity directly. The partial derivative of the left side with respect to H , at $(z, H) = (0, 1)$, is -32 . Thus the formal implicit-function theorem singles out the unique branch with $H(0) = 1$. In particular, no ambiguous choice of an algebraic root is hidden in the generating-function description.

The symbolic verification program checks both the elimination and the series identity through degree 20. The coefficient recurrence (2.3) is used for that series check, so it does not merely substitute one rearranged symbolic expression into another.

7 A boundary test for periodic-coefficient arguments

7.1 What the logarithmic method actually requires

The odd-prime proof used two different ingredients: the equality $c_{pj} = c_j$ and the much stronger quadratic cancellation in Lemma 3.3. The first does not imply the second.

More generally, let $\Phi \in 1 + xR_p[[x]]$, and suppose

$$\log \Phi(x) = \sum_{j \geq 1} c_j x^j / j, \quad c_j \in R_p, \quad c_{pj} = c_j.$$

Define $A(n) = [x^n] \Phi(x)^n$ and D_p as in (3.4). The factorization and the vanishing linear term still hold. For odd p , all terms of degree at least two in the exponential have valuation at least $2v_p(N)$, so they yield a quadratic tower. A cubic tower requires an additional estimate on the coefficients of D_p^2 .

At the first layer and for $p \geq 5$, this gives a particularly transparent necessary and sufficient residue condition:

$$A(p) - A(1) \equiv -\frac{p^2}{2} \sum_{j=1}^{p-1} \frac{c_j c_{p-j}}{j^2} \pmod{p^3}. \quad (7.1)$$

Indeed, the quadratic coefficient at degree p is unaffected by positive powers of x^p from $\Phi(x^p)$, and $1/(j(p-j)) \equiv -1/j^2 \pmod{p}$. This proves the formula without any periodicity assumption beyond $c_{pj} = c_j$.

7.2 A spacing-three counterexample

Replace the even-occupancy restriction by occupancy divisible by 3:

$$b(n) = [x^n](1-x)^{-n}(1-x^3)^{-n}. \quad (7.2)$$

Its logarithmic coefficients are $c_j = 1 + 3\mathbf{1}_{3|j}$. They have period three and satisfy $c_{5j} = c_j$. Nevertheless,

$$\begin{aligned} b(1) &= 1, \\ b(5) &= \binom{9}{5} + 5 \binom{6}{2} = 126 + 75 = 201, \\ b(5) - b(1) &= 200 \not\equiv 0 \pmod{125}. \end{aligned}$$

The obstruction in (7.1) is exactly

$$\sum_{j=1}^4 \frac{c_j c_{5-j}}{j^2} = 1 + \frac{4}{4} + \frac{4}{9} + \frac{1}{16} = \frac{361}{144} \equiv 4 \pmod{5}. \quad (7.3)$$

Equation (7.1) predicts the residue 75 modulo 125, which agrees with $200 \pmod{125}$.

Thus the success of period two is structural, not an automatic consequence of rationality, periodicity, or Frobenius invariance. In the period-two proof, the weighted square sum collapses to two ordinary reciprocal-square sums. That special collapse is absent here.

7.3 Scope of the literature correction

Müller’s arXiv version 1 of *Wolstenholme Type Congruences and Framing of Rational 2-Functions* states a general periodic weighted-harmonic bound in Theorems 1.2 and 6.2 and a corresponding precise cubic framing bound in Theorem 1.1 [7]. The value (7.3) contradicts the printed harmonic bound at period 3 and prime 5.

The coefficient counterexample also fits the printed number-field hypotheses, not merely a superficial formal analogy. Take $K = \mathbb{Q}(\sqrt{-3})$, whose discriminant is -3 , and

$$V(x) = \frac{x}{1-x} + \frac{3x^3}{1-x^3}.$$

At every unramified rational prime $p \neq 3$, its rational coefficients satisfy $c_{mp^r} = c_{mp^{r-1}}$ exactly. Thus it is a rational 2-function in the convention of that preprint. Formula (5.3) there identifies the coefficient of its plus-framing with parameter one as $[x^n] \exp(n \int V) = b(n)$, where the formal integration sends x^j to x^j/j . Prime 5 is unramified and does not divide the period. The precise bound in Theorem 1.1 would therefore give divisibility by 125, contrary to $b(5) - b(1) = 200$.

A different period-four counterexample is already documented by the same repository that supplied the prior odd-prime proof [4]. Accordingly, the discovery of a problem in the general framing statement is not claimed as new here. The spacing-three example is a compact independent boundary witness. The criticism is specifically of the printed uniform bounds in version 1. A single exceptional prime does not refute a weaker assertion that permits an unspecified finite set of additional exceptional primes, and it says nothing by itself about unrelated framing results. None of the proofs in this article uses the disputed statements.

8 Asymptotic expansion of the basket sequence

The leading asymptotic is recorded in the OEIS entry, attributed to Kotesovec [1]. Here we derive it together with two explicit correction terms and an all-order coefficient formula. The method is the standard saddle-point approach to large powers of generating functions [8, Chapter VIII].

8.1 The saddle and exponential growth

For this section only, put

$$\Phi(x) = \frac{1}{(1-x)(1-x^2)}.$$

The positive saddle $\tau \in (0, 1)$ satisfies

$$\tau \frac{\Phi'(\tau)}{\Phi(\tau)} = 1.$$

Since $x\Phi'/\Phi = x/(1-x) + 2x^2/(1-x^2)$, this is equivalent to $4\tau^2 + \tau - 1 = 0$. Hence

$$\tau = \frac{\sqrt{17}-1}{8}, \quad \rho = \frac{\tau}{\Phi(\tau)} = \frac{51\sqrt{17}-107}{512}, \quad \lambda = \rho^{-1} = \frac{107+51\sqrt{17}}{64}. \quad (8.1)$$

Numerically, $\lambda = 4.9574747954140732506\dots$. Define the logarithmic cumulants

$$\kappa_j = \left(x \frac{d}{dx} \right)^j \log \Phi(x) \Big|_{x=\tau}. \quad (8.2)$$

Then $\kappa_1 = 1$ and

$$\begin{aligned}\kappa_2 &= \frac{51 - 5\sqrt{17}}{16}, & \kappa_3 &= \frac{435 - 69\sqrt{17}}{32}, \\ \kappa_4 &= \frac{10731 - 2125\sqrt{17}}{128}, & \kappa_5 &= \frac{91545 - 20175\sqrt{17}}{128}, \\ \kappa_6 &= \frac{4008567 - 929185\sqrt{17}}{512}.\end{aligned}\tag{8.3}$$

Theorem 8.1 (Two correction terms). *As $n \rightarrow \infty$ through positive integers,*

$$a(n) = \frac{\lambda^n}{\sqrt{2\pi\kappa_2 n}} \left(1 + \frac{c_1}{n} + \frac{c_2}{n^2} + O(n^{-3}) \right),\tag{8.4}$$

where

$$c_1 = \frac{-1683 + 95\sqrt{17}}{9248}, \quad c_2 = \frac{119821 - 28605\sqrt{17}}{5030912}.\tag{8.5}$$

The leading amplitude is equivalently

$$\frac{1}{\sqrt{2\pi\kappa_2}} = \frac{\sqrt{3 + 5/\sqrt{17}}}{4\sqrt{\pi}}.\tag{8.6}$$

8.2 Derivation, including control of the remainder

Cauchy's coefficient formula on the circle $|x| = \tau$ gives

$$a(n) = \frac{\lambda^n}{2\pi} \int_{-\pi}^{\pi} \exp \left(n \left\{ \log \frac{\Phi(\tau e^{i\theta})}{\Phi(\tau)} - i\theta \right\} \right) d\theta.\tag{8.7}$$

The coefficients of Φ are positive and its support contains 0 and 1. Equality in the triangle inequality $|\Phi(\tau e^{i\theta})| \leq \Phi(\tau)$ therefore requires $e^{i\theta} = 1$. On every closed subinterval bounded away from zero, the integrand is exponentially smaller than the central contribution.

Near zero, choose the analytic logarithm agreeing with the real logarithm at τ . The saddle condition cancels the linear term, and the exponent inside braces is

$$-\frac{\kappa_2 \theta^2}{2} + \sum_{k \geq 3} \frac{\kappa_k (i\theta)^k}{k!}.\tag{8.8}$$

In particular, its real part is at most $-c\theta^2$ for sufficiently small $|\theta|$ and some $c > 0$.

Set $u = \sqrt{n\kappa_2} \theta$. The central integral becomes a Gaussian integral multiplied by

$$\exp \left(\sum_{k \geq 3} \frac{\kappa_k i^k u^k}{k! \kappa_2^{k/2}} n^{1-k/2} \right).$$

Expanding this expression and integrating term by term is justified at any fixed order by analytic Taylor remainder estimates and Gaussian domination. For specificity, split a small fixed central interval at $|\theta| = n^{-2/5}$. Outside that shrinking interval the bound $e^{-cn\theta^2}$ is exponentially small in $n^{1/5}$. Inside it, Taylor expansion to sufficiently high finite degree has a uniform remainder; rescaling and integrating its polynomial bounds against a slightly wider Gaussian gives the stated power remainder. Odd powers have zero symmetric Gaussian moment. Carrying the expansion two orders further than the last retained even power bounds the relative remainder by $O(n^{-3})$ in (8.4). The same argument works to every fixed order.

Using

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} u^{2\ell} e^{-u^2/2} du = (2\ell - 1)!!,$$

one obtains

$$\begin{aligned} c_1 &= \frac{\kappa_4}{8\kappa_2^2} - \frac{5\kappa_3^2}{24\kappa_2^3}, \\ c_2 &= -\frac{\kappa_6}{48\kappa_2^3} + \frac{7\kappa_3\kappa_5}{48\kappa_2^4} + \frac{35\kappa_4^2}{384\kappa_2^4} \\ &\quad - \frac{35\kappa_3^2\kappa_4}{64\kappa_2^5} + \frac{385\kappa_3^4}{1152\kappa_2^6}. \end{aligned} \tag{8.9}$$

Substitution of (8.3) simplifies to (8.5); elementary rationalization gives (8.6). This proves the theorem.

8.3 An explicit all-order coefficient formula

The same calculation gives a finite expression for every correction coefficient. For $j \geq 0$, sum over nonnegative integers $m_3, \dots, m_{2j+2}$ satisfying

$$\sum_{k=3}^{2j+2} (k-2)m_k = 2j, \quad K = \sum_{k=3}^{2j+2} km_k.$$

The integer K is even. Then

$$c_j = \sum (-1)^{K/2} (K-1)!! \kappa_2^{-K/2} \prod_{k=3}^{2j+2} \frac{\kappa_k^{m_k}}{m_k! (k!)^{m_k}}. \tag{8.10}$$

For $j = 0$, the empty product and $(-1)!!$ are both 1, giving $c_0 = 1$. The constraint comes from collecting the exponent of $n^{-1/2}$ in the expanded exponential; the double factorial comes from its Gaussian moment. This is a direct finite enumeration formula, not a recurrence with unknown initial correction terms.

Numerically, $c_1 \approx -0.1396307272454933$ and $c_2 \approx 0.0003736029529450764$. The cancellation in c_2 is substantial, so exact symbolic simplification is preferable to low-precision arithmetic. The supplied numerical checks use 90 decimal digits. For illustration, the signed relative error of the approximation through c_2 is about $-9.6264 \cdot 10^{-9}$ at $n = 100$ and $-6.1254 \cdot 10^{-10}$ at $n = 250$. Those numerical comparisons are checks, not the proof of the remainder.

9 Reproducible computation and proof audit

9.1 What the programs compute

The archive separates exact integer checks from optional symbolic and high-precision checks. The program `verify.py` uses only the Python standard library and implements (2.3); it cross-checks against (2.1) and (2.2). The program `binary_checks.py` tests the complete leading-defect formula, the harmonic-unit lemma, and both common-denominator assertions. The program `symbolic_checks.py` uses SymPy and mpmath for elimination, Gaussian coefficient enumeration, and asymptotic comparisons.

The recorded run produced the following exact checks. Each count is a finite coverage statement, not an extrapolation to all indices.

Check	Cases
Initial target terms	15
Target recurrence versus positive binomial sum	208
Signed parameter recurrence versus binomial sum	1,519
Target odd-prime congruences	420
Two-parameter odd-prime congruences	3,136
Odd-prime quadratic-tail tests	918
Original Möbius/orbit checks	300
Original boundary witnesses	5
Complete binary leading-defect formula	5,808
Binary first-lift congruence	1,936
Exact binary harmonic-unit normalization	400
Target denominator-12 and orbit divisibility	1,000
General denominator-24 / conditional denominator-12	4,900
Binary sharpness and minimal-denominator witnesses	290

All these checks passed. The binary leading-defect grid covers $-5 \leq \alpha, \beta \leq 5$, odd $1 \leq m \leq 15$, and $2 \leq r \leq 7$, with $m2^r \leq 2000$. The target odd-prime run uses odd primes at most 97, multipliers at most 12 coprime to the prime, and indices at most 3000. The general odd-prime grid uses parameters from -3 to 3 , primes $3, 5, 7, 11, 13$, and indices at most 350.

The symbolic run additionally verifies the logarithmic derivative, the quartic resultant, its parametrization, its power series through degree 20, the saddle point, the growth constant, the leading amplitude, and the Gaussian formulas for c_1 and c_2 . These tests used Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0 in the recorded environment; the exact scripts are not tied to those library versions.

9.2 Reproduction commands and output data

From the archive directory, run

```
python verify.py
python binary_checks.py
python -m pip install -r requirements-optional.txt
python symbolic_checks.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Only the third and fourth commands need optional Python packages. No network access is used by any of the checking programs themselves. The integer checker accepts command-line coverage parameters; see `python verify.py -help`.

The CSV files contain target terms through 1000, the exact prime-power valuation checks, the full binary parameter grid, primitive-orbit and Möbius values, and asymptotic comparisons. The reports preserve the actual check counts. There are no randomized tests and no unrecorded dependence on an external computer-algebra session.

9.3 Independent points of possible failure

Several interfaces deserve explicit scrutiny in any independent review. First, a recurrence in the coefficient index must not be mistaken for a recurrence in the diagonal index. Second, the odd-prime exponential has a vanishing linear term, but the binary one does not; the latter is handled by reflection, not by reusing the former proof. Third, the $1/2$ in formal integration by parts at the binary prime is retained throughout and is responsible for one of the losses.

Fourth, the Möbius grouping is performed at the exact prime-power exponent of n , not merely at a prime dividing n . Finally, the finer binary valuation table is not used to justify any of the uniform bounds.

The article uses only elementary formal series, local rational arithmetic, coefficient extraction, and the stated saddle-point analysis. It does not invoke the disputed periodic framing bound or any unverified asymptotic recurrence from an OEIS comment.

10 Further questions after the binary completion

The uniform small-prime completion is proved, but it does not determine all exceptional extra divisibility. For the basket sequence, the observed valuations at $n = 2^r$ are

r	1	2	3	4	5	6	7	8
$v_2(a(2^r) - a(2^{r-1}))$	2	4	8	13	16	19	22	25

The recorded exact computation continues this pattern through $r = 15$.

Conjecture 10.1 (A finer target valuation, not used above). For every $r \geq 4$,

$$v_2(a(2^r) - a(2^{r-1})) = 3r + 1.$$

This is not a consequence of Theorem 1.2, whose leading residue vanishes for $(\alpha, \beta) = (2, 1)$. Proving it would require following the reflection calculation several binary digits further, including the next residues of $K(t)$ and additional exponential terms. The statement for a fixed odd multiplier is more varied: at $m = 5$ the values from $r = 4$ onward in the checked range are $3r + 3$, while at $m = 7$ they are $3r + 2$.

A second question is to characterize the periodic logarithmic coefficient sequences for which the cubic tail estimate holds. Formula (7.1) gives an immediate first-level obstruction, but vanishing at the first level alone does not establish all higher levels. The spacing-three example shows why a condition stronger than periodicity and Frobenius invariance is needed. A useful classification would separate identities forced by symmetry from congruences that depend on the prime.

A third question is to refine the common denominator for a fixed parameter pair. The bounds 24 and 12 are optimal uniformly over the stated classes, but special pairs can have smaller denominators. The exact first binary residue already detects one such improvement. An analogous explicit ternary residue could identify further parameter-dependent reductions.

These questions are distinct from the completed claims. In particular, the remaining uncertainty about the finer valuation pattern does not limit the proof of the universal denominator-12 theorem.

11 Conclusion

The initial OEIS label led to a useful but necessary correction of the research target: a prior odd-prime proof candidate already existed. The binary completion proved here uses a different mechanism, converting a linear exponential contribution into a quadratic one by reflection and then extracting an extra index factor by formal integration by parts. An exact odd-harmonic normalization identifies the first possible binary bit, not just a lower bound.

For A348410 the completed all-prime estimate is naturally encoded by the single integer 12. It clears every denominator of the cubic Möbius transform and is forced by the first few terms. The resulting primitive cyclic-word divisibility gives a combinatorial expression of the same arithmetic fact. The archive contains the full proofs and reproducible checks; priority beyond the audited sources and external validation of the new binary argument remain matters for further review.

A A minimal exact evaluator

The following code is the computational core; the full scripts add input validation, independent formula checks, report generation, and CSV output.

```
def coefficient_diagonal(n: int, alpha: int = 2,
                        beta: int = 1) -> int:
    if n < 0:
        raise ValueError("n must be nonnegative")
    f_minus_two, f_minus_one = 0, 1
    for k in range(1, n + 1):
        numerator = n * (alpha - beta) * f_minus_one
        numerator += (n * (alpha + beta) + k - 2) * f_minus_two
        current, remainder = divmod(numerator, k)
        if remainder:
            raise ArithmeticError("coefficient division was not exact")
        f_minus_two, f_minus_one = f_minus_one, current
    return f_minus_one
```

The running names refer to the two previously available coefficients at the start of each iteration. At $n = 0$ the function correctly returns 1.

B Notation and dependency map

Symbol	Meaning
$A_{\alpha,\beta}(n)$	Integer-parameter coefficient diagonal in (1.4).
$a(n)$	The specialization $A_{2,1}(n)$, OEIS A348410.
R_p	Rationals with denominator prime to p .
D_p	Reduced odd-prime logarithm in (3.4).
$H_p(s)$	Reduced reciprocal-square sum; not the generating function.
$h(T)$	Sum of reciprocals of the positive odd integers less than T .
U, K, V	Binary series in (4.1) and (4.4).
$B(n)$	Cubic Möbius transform $n^{-3} \sum_{d n} \mu(d) a(n/d)$.
P_n	Primitive cyclic-word count, equal to $n^2 B(n)$.
$H(z)$	Ordinary generating function of $a(n)$.
κ_j	Saddle cumulants in (8.2).

The proof dependencies are short. Lemmas 3.1–3.3 imply Theorem 3.4. Lemma 4.1, reflection, and (4.12) imply Theorem 1.2. These two theorems plus the first binary lift imply Theorem 1.3. Grouping Möbius divisors then proves Theorem 1.4; periodic word decomposition gives Corollary 1.5. The generating-function and asymptotic sections are independent checks and complementary structure, not premises of the arithmetic proofs.

References

- [1] OEIS Foundation Inc., *A348410: Number of nonnegative integer solutions to $n = \sum_{i=1}^n (a_i + b_i)$, with b_i even*. Includes P. Bala’s 21 February 2022 coefficient formulas and supercongruence conjecture, and V. Kotesovec’s leading asymptotic. <https://oeis.org/A348410>. Inspected 19 September 2026.
- [2] OEIS Foundation Inc., *A352373*. Related parameterized coefficient-power congruence. <https://oeis.org/A352373>. Inspected 19 September 2026.
- [3] GitHub repository `rbajaj5/a183068-supercongruence`, *A cubic tower for a two-parameter coefficient family*, `related-results/CoefficientFramingCubicTower.md`. Public proof candidate; pinned repository snapshot 3085b46fdbce (full hash in the linked URL). [Permanent link to the inspected note](#). Inspected 19 September 2026. The source states no binary assertion.
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